PROGRESS REPORT

MC-ESO

Multi-Channel Epidemic Spread Optimizer

A population-based black-box optimizer inspired by
multi-route epidemic transmission

M1 Kosei Matsuzaki

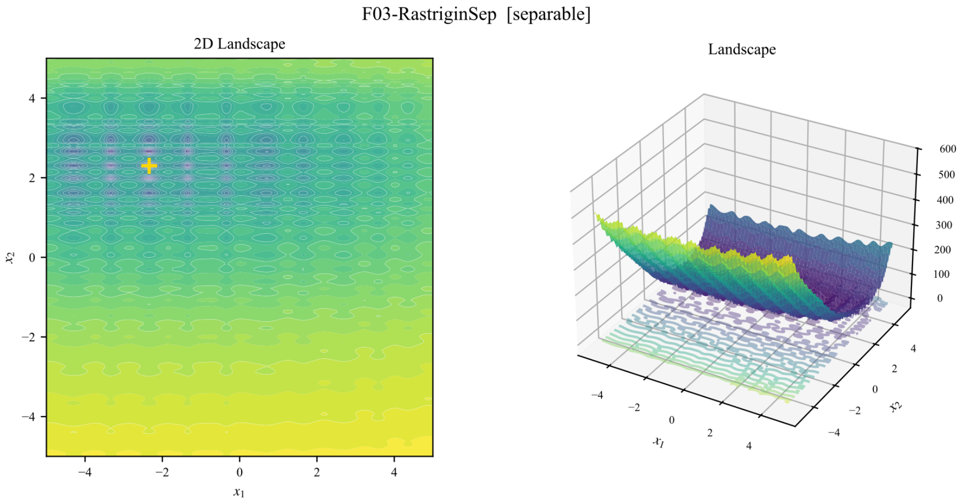
Outline

1

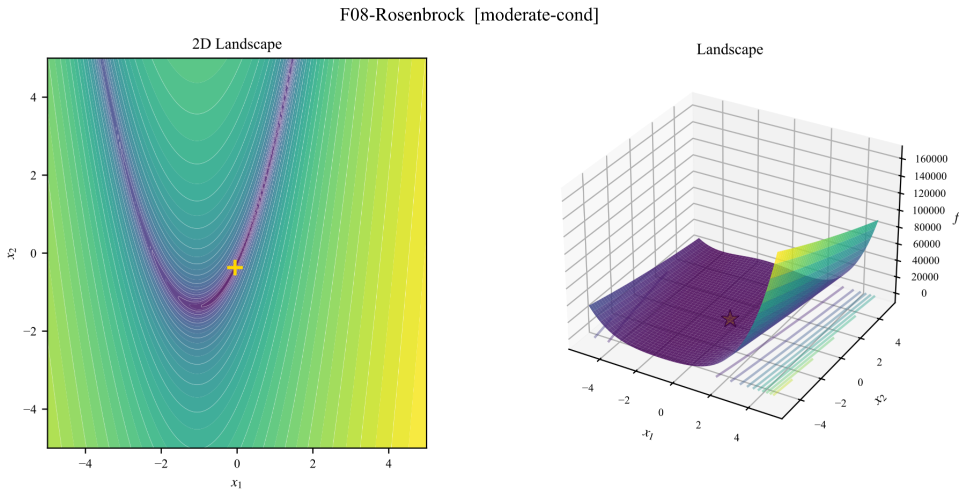
Background

The shared weakness of existing methods, and what motivated this work

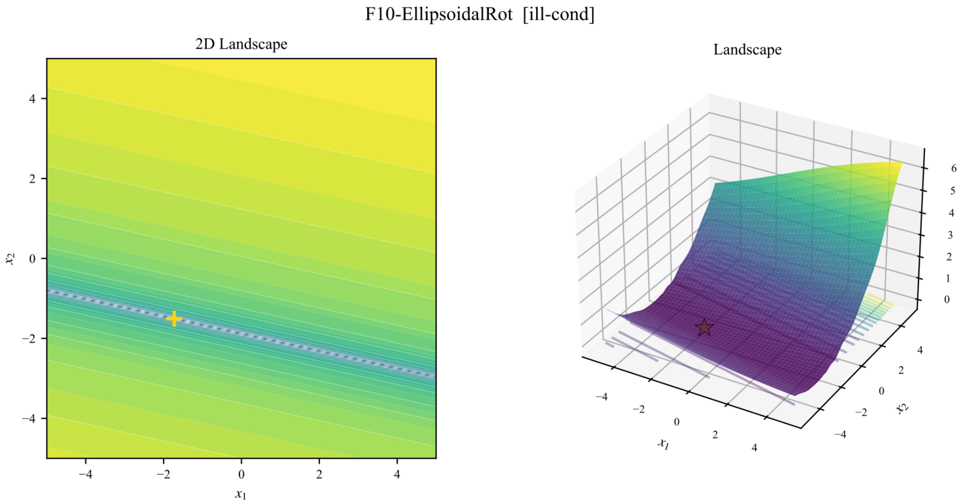
Black-box functions



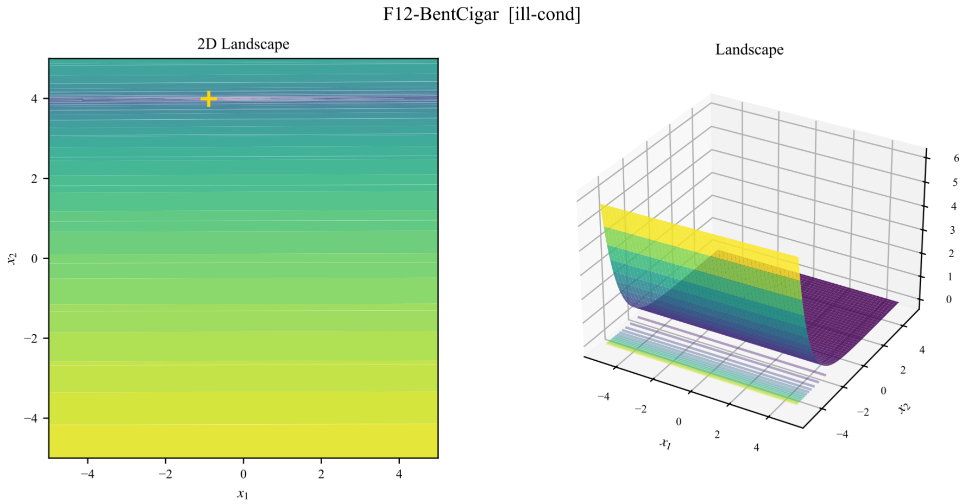
Multimodal



Narrow curved valley



Ill-conditioned

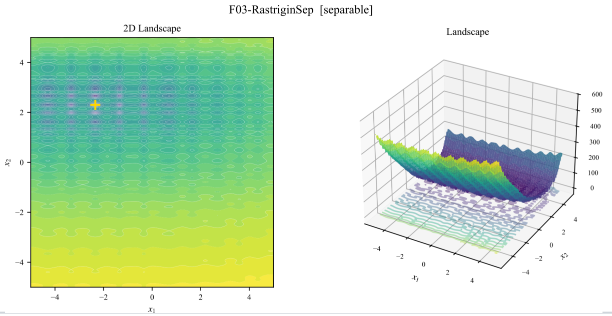


Extreme anisotropy

Each existing method fails on a class of problems

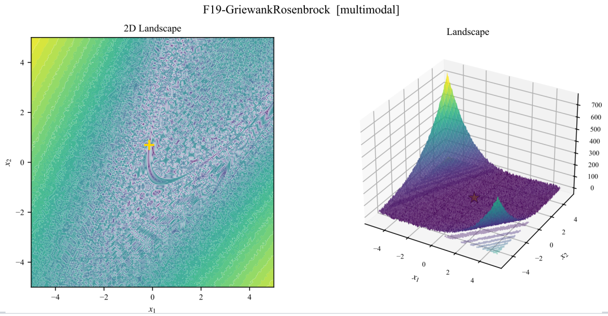
CMA-ES

Multimodal traps



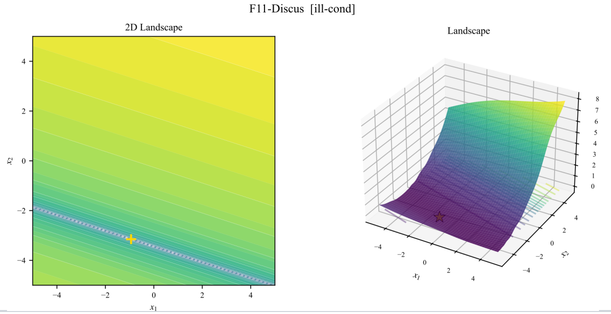
DE

Weak structure



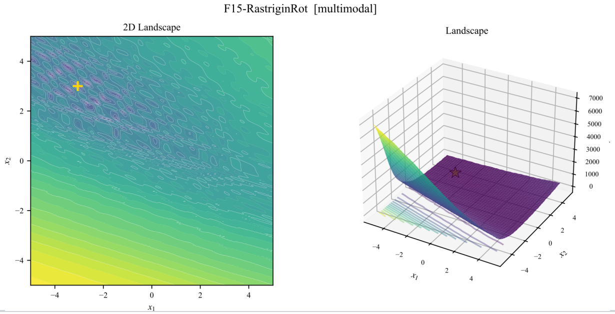
PSO

Ill-conditioned



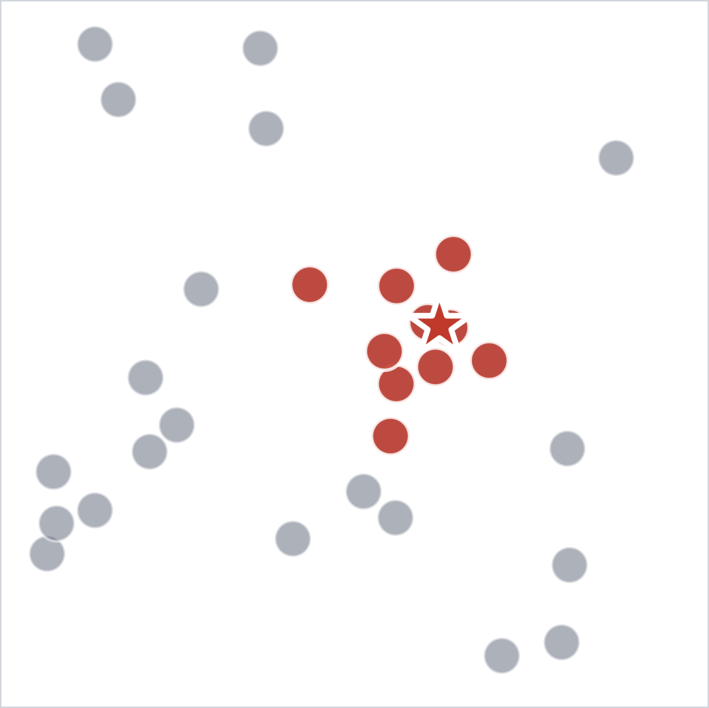
SaVOA

Escape or refine, not both



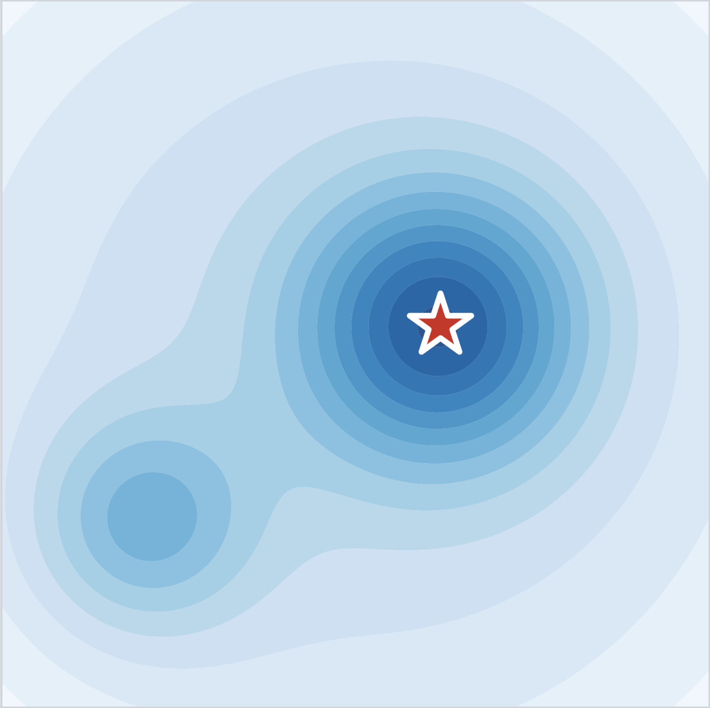
Epidemic spread and optimization share the same structure

Epidemic spread



Epidemic spread	Optimization
Viral particle	Sample point
Host-density map	Objective landscape
Outbreak in dense area	Finding the optimum
Multi-site outbreaks	Multi-basin diversity

Optimization



Related work — virus- / epidemic-inspired optimization

Each prior method models a single transmission rule

Year	Method	Mechanism / how it generates samples
2014	**VOA (Liang & Juarez) [1]**	Strong/common virus classes + antivirus pruning — one Gaussian channel
2020	**SaVOA (Liang & Juarez) [2]**	Self-adaptive VOA, single channel
2020	**CVOA (Martínez-Álvarez et al.) [3]**	COVID-19 infection→isolation cycle, single rule
2022	**EOSA (Oyelade et al.) [4]**	Ebola SEIR compartments (S/E/I/R)
2022	**ACVO (Hashemi et al.) [5]**	Containment-style search
2025	**VDO (Virus Diffusion) [6]**	Herpes life-cycle: tropism + replication + diffusion + latency

Deep-dive — SaVOA (Liang & Juarez)

Soft Computing, 2020 [2] · Self-adaptive variant of VOA

1

Two Virus Classes

<u>Strong exploit</u> & <u>Common explore</u>
— same Gaussian, different radius.

2

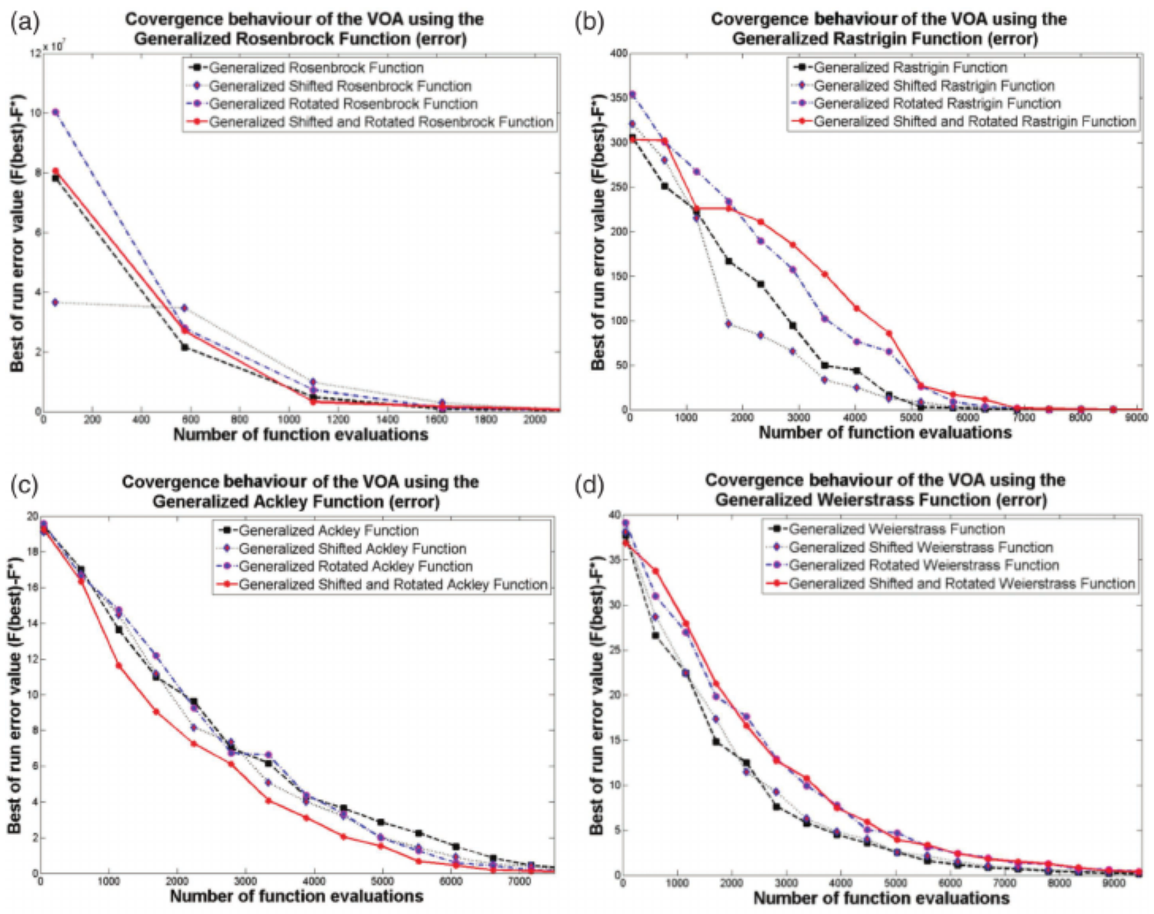
Replicate + Antivirus

Parents spawn Gaussians.
Antivirus culls the worst.

3

Self-adapt Rates

Rates adjust online — 5 HPs → 1.



Deep-dive — VDO Virus Diffusion Optimizer

arXiv, 2025 [6] · Most recent virus-inspired method — herpes life-cycle with four stages

1

Tropism Exploration + Replication

Burst around best on dims that beat it, with decaying step.

2

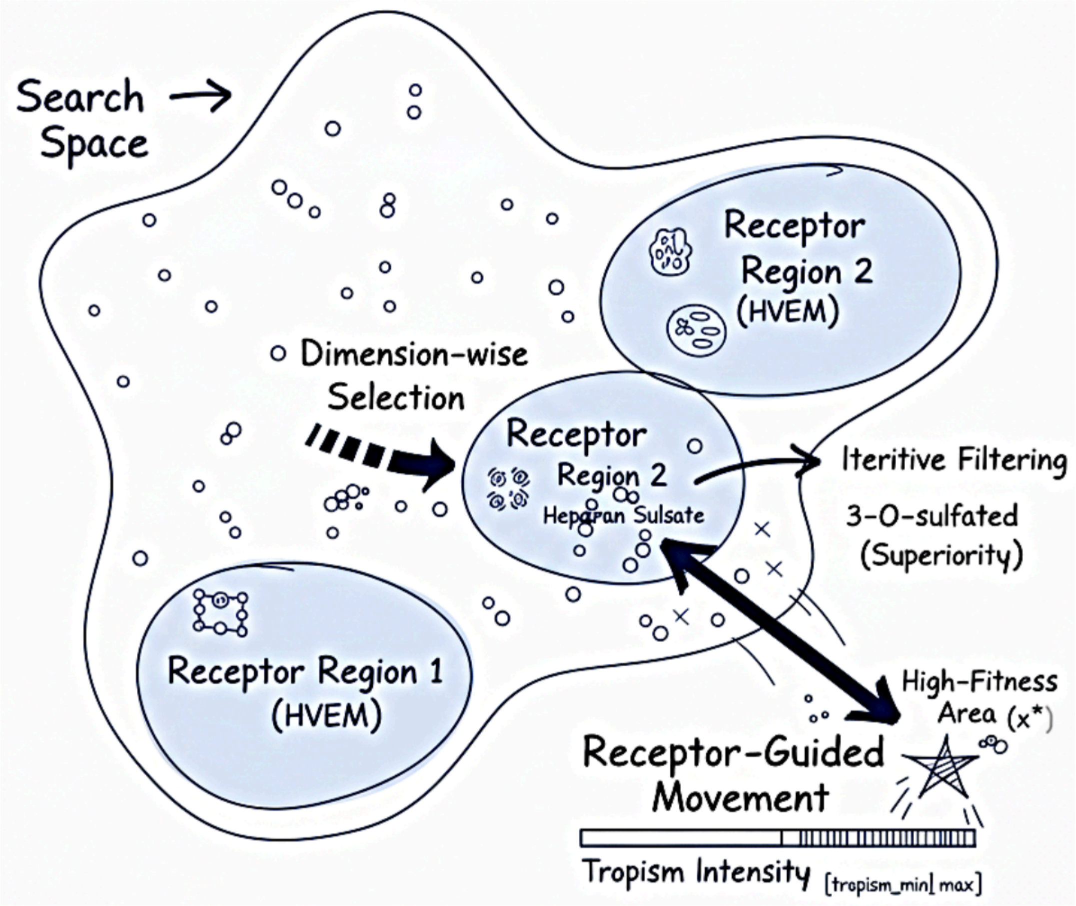
Diffusion (local + Lévy)

Small local moves + occasional Lévy jumps.

3

Latency reactivation

Re-inject dormant solutions when stuck — auto-restart.



2

Purpose & Significance

What we ask, and why it matters

Goal & contributions

GOAL

Build a black-box optimizer that performs well on every function

Novelty	First multi-route, epidemic-inspired optimizer
Mechanism	Channel mix + population control + adaptive σ
Outcome	≥75% SR on every BBOB category

Result preview — overall and per-category SR@1e-4

35 functions, 30 runs each

MC-ESO	DE	SaVOA	PSO	CMA-ES
90.9%	**88.7%**	**75.7%**	**75.3%**	**63.0%**

Category	MC-ESO	DE	SaVOA	PSO	CMA-ES
Separable	**99%**	**99%**	81%	97%	69%
Moderate cond.	**100%**	98%	**100%**	**100%**	94%
Ill-conditioned	97%	97%	49%	40%	**100%**
Multi-optima	**100%**	98%	**100%**	**100%**	97%
Multimodal	**98%**	97%	69%	82%	45%
Weak structure	**83%**	67%	65%	49%	47%
Deceptive (2D)	75%	78%	**79%**	74%	30%

3

Algorithm Sketch

3 transmission channels → 3 population mechanisms → adaptive σ

One generation of MC-ESO

5 steps per generation

REPRODUCE

1	**Spawn**	Three transmission channels	Contact · Droplet · Airborne
2	**Evaluate**	—	Single fitness call per child

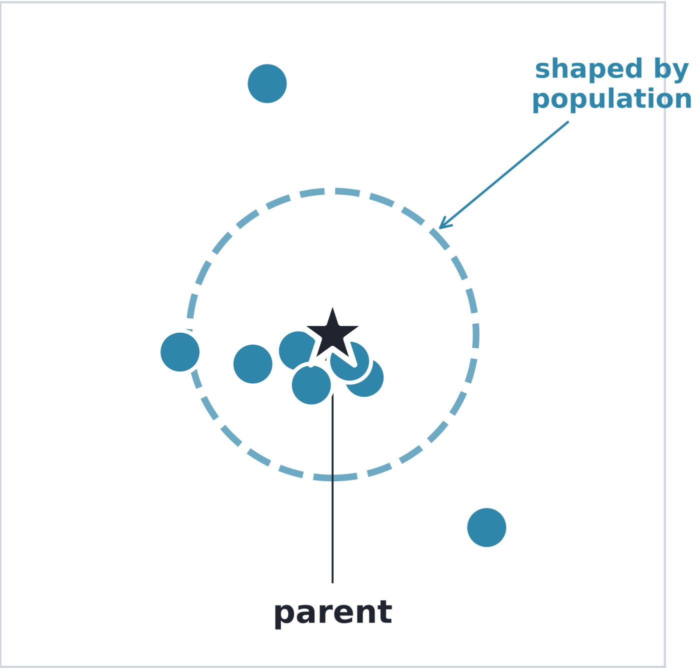
SELECT & ADAPT

3	**Update strain pool**	Strain coexistence	Keep up to 6 spatially-separated elites
4	**Select population**	Host competition	Parents-plus-children greedy with rollback
5	**Restart + σ adapt**	Spillover + drilling mode	Re-seed or basin switch, plus a tighter σ in drilling

Three transmission channels

Close-contact

p = 30 %

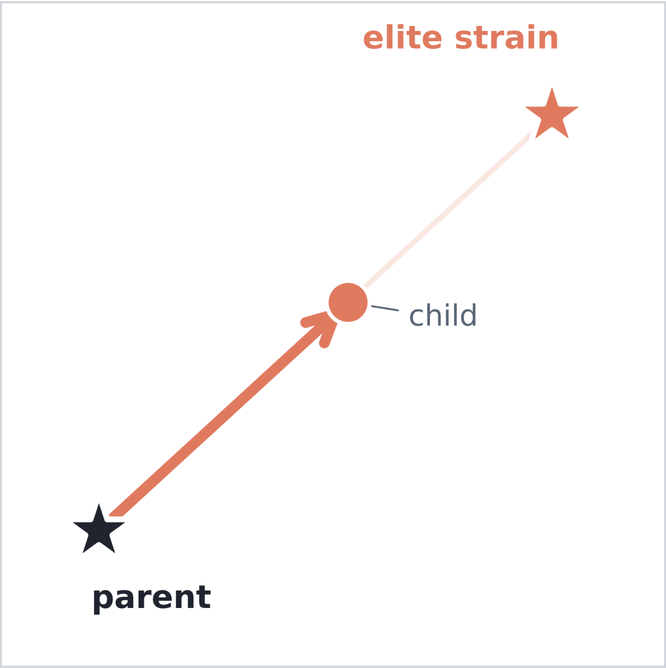


Mimics: Person-to-person touch

Local Gaussian shaped by C_pop

Droplet

p = 40 %

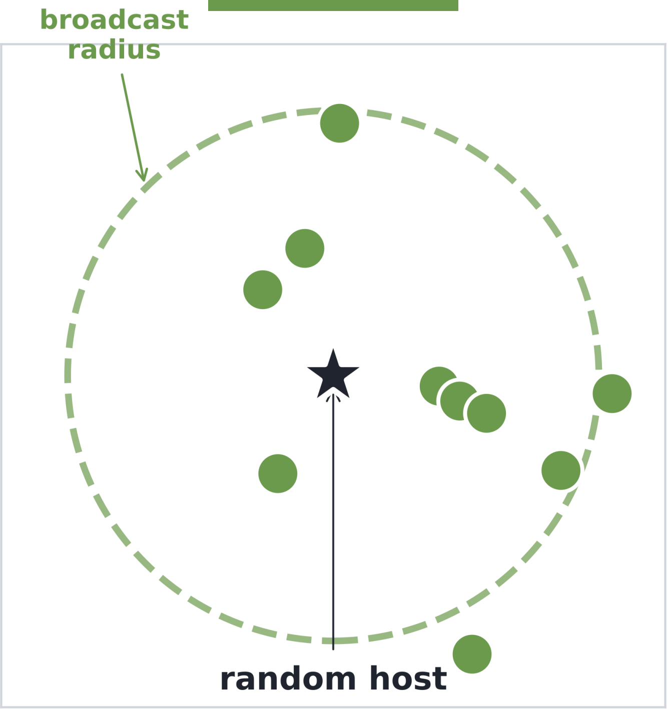


Mimics: Pseudo flow of people

DE/current-to-best with strain pull

Airborne

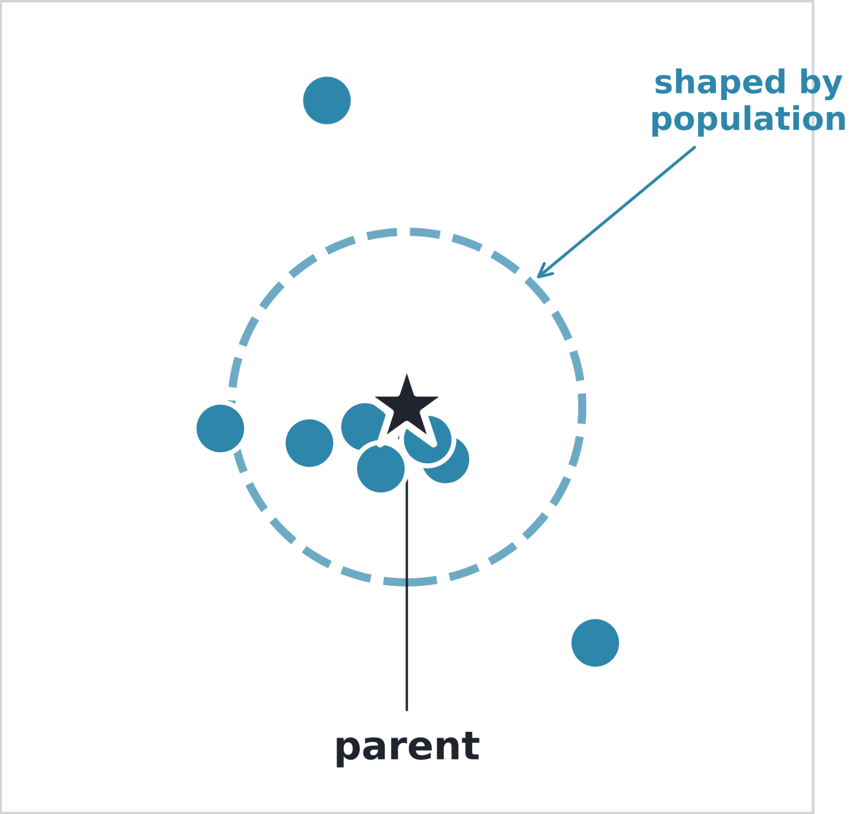
p = 30 %



Mimics: Aerosols drift far

Wide random spread, off in drilling

Channel — Close-contact



Gaussian cloud around the parent, shaped by the population's spread

Close-contact

Rotation-aware local refinement

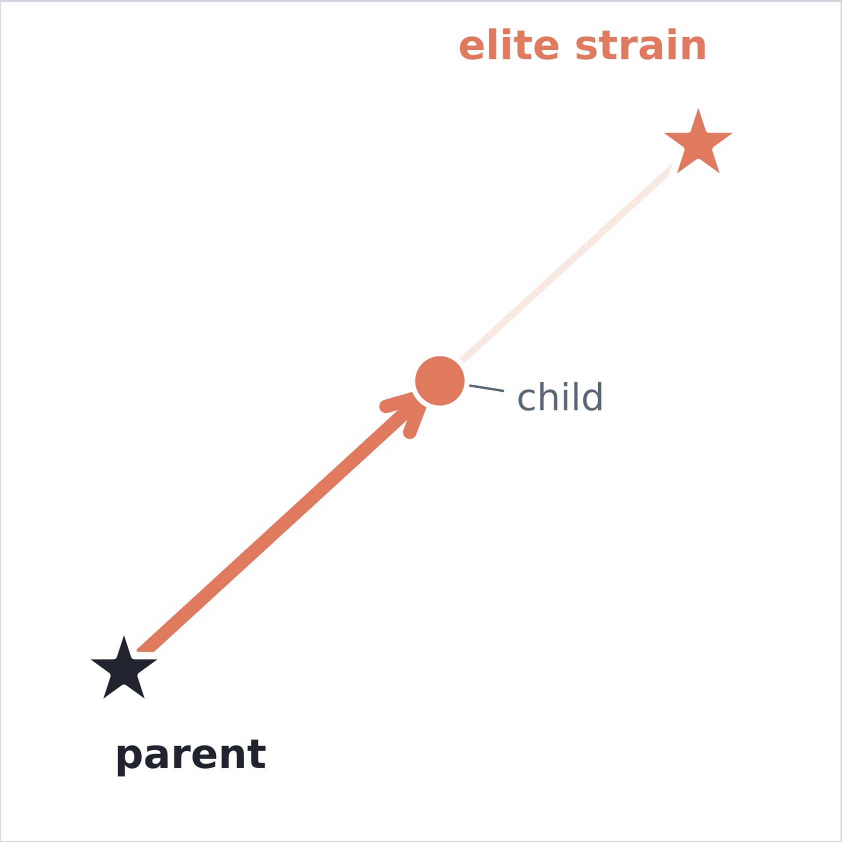
Formula

$$\boldsymbol{x}_p + \mathcal{N}(0,\ \sigma_i^2\, \boldsymbol{C}_{\mathrm{pop}})$$

How it works

1. Pick a parent with its own adaptive σ
2. Add Gaussian noise shaped by the population covariance
3. Covariance is taken empirically, with no history

Channel — Droplet



Parent pulled partway toward an elite strain

Droplet

Strain-guided DE move

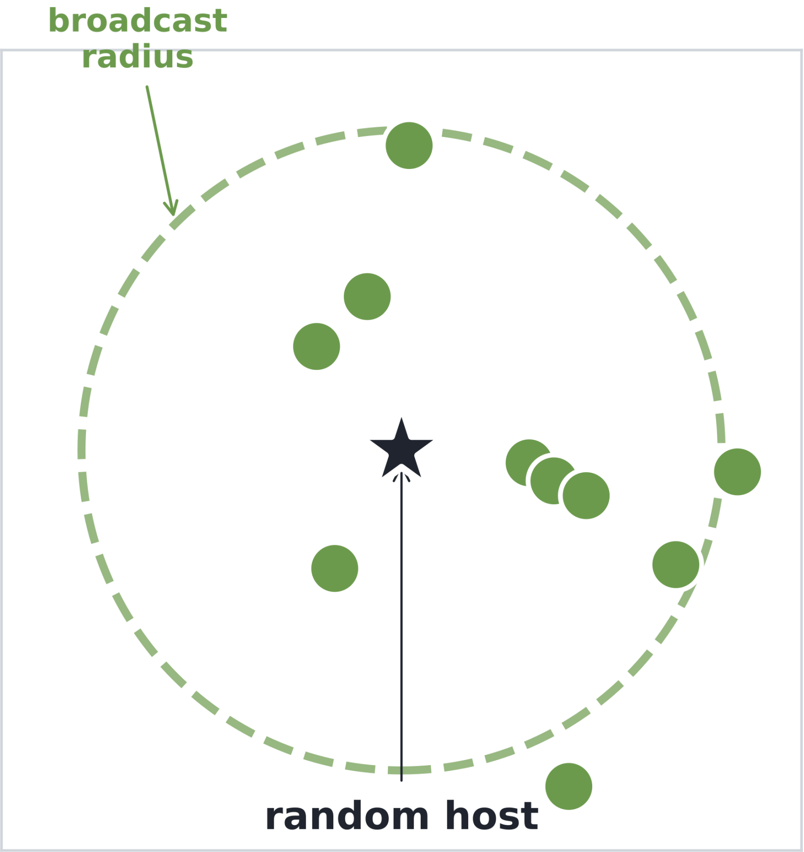
Formula

$$\boldsymbol{x}_p + F(\boldsymbol{x}_{\mathrm{strain}} - \boldsymbol{x}_p) + F(\boldsymbol{x}_a - \boldsymbol{x}_b)$$

How it works

1. Pull toward a niched elite (current-to-best style)
2. Add a difference vector between two random peers
3. Mix with the parent via binomial crossover (rate 0.9)

Channel — Airborne



Wide scatter around a random host

Airborne

Wide broadcast — escape when stuck

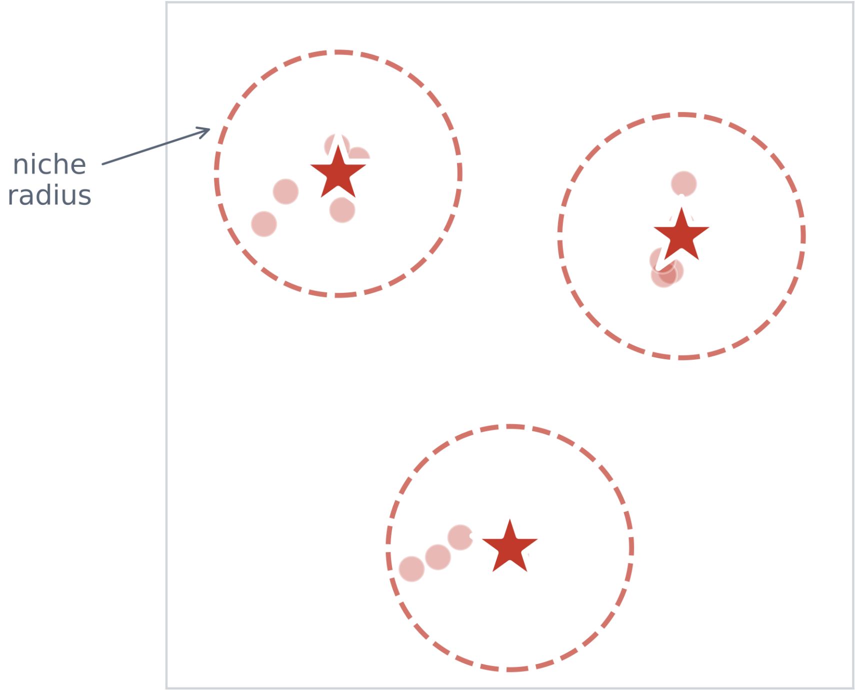
Formula

$$\boldsymbol{x}_{\text{rand}} + \mathcal{N}(0,\ \sigma_{\text{air}}\,\boldsymbol{I})$$

How it works

1. Sample around a randomly chosen host
2. The broadcast radius widens as diversity drops
3. Switched off in drilling mode (when σ becomes very small)

Population mechanism — Strain coexistence



Three coexisting strains with niche radii

Strain coexistence

Protect up to 6 spaced elites

Activates when

A fit child lands outside every existing niche.

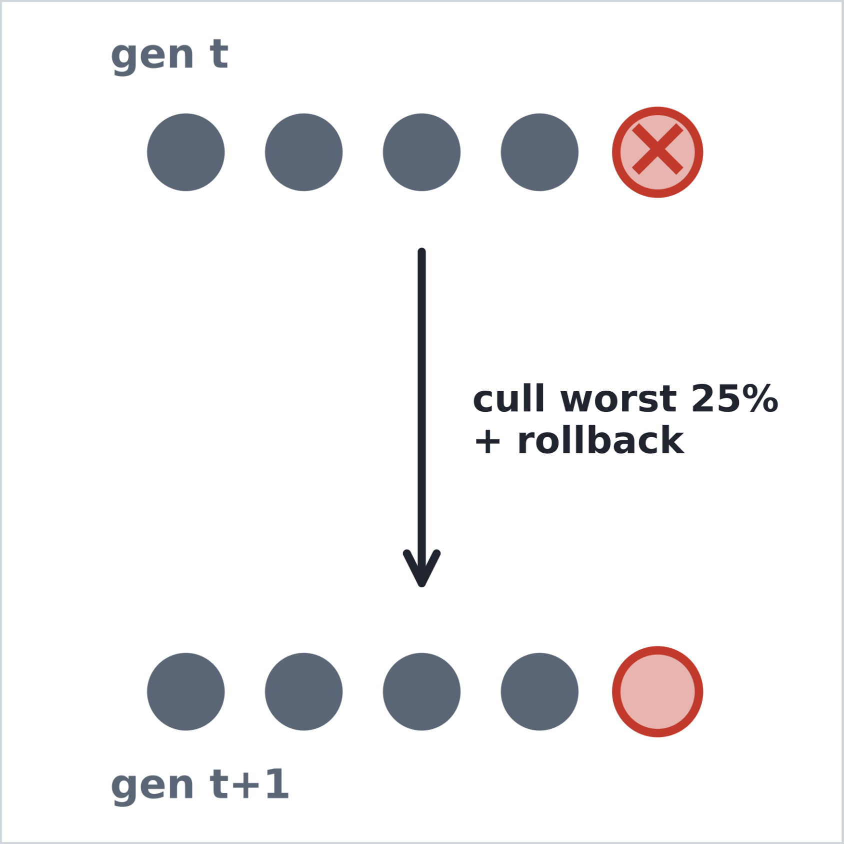
What happens

Register it as a new strain.
Cap at 6 strains, with niches spaced by roughly a tenth of the search range.

Why it helps

Droplet can pull toward ANY strain — diversity persists.

Population mechanism — Host competition



Worst 25% removed, losing child rolled back

Host competition

Kill the worst 25%, then rollback worse children

Activates when

End of generation — all children evaluated.

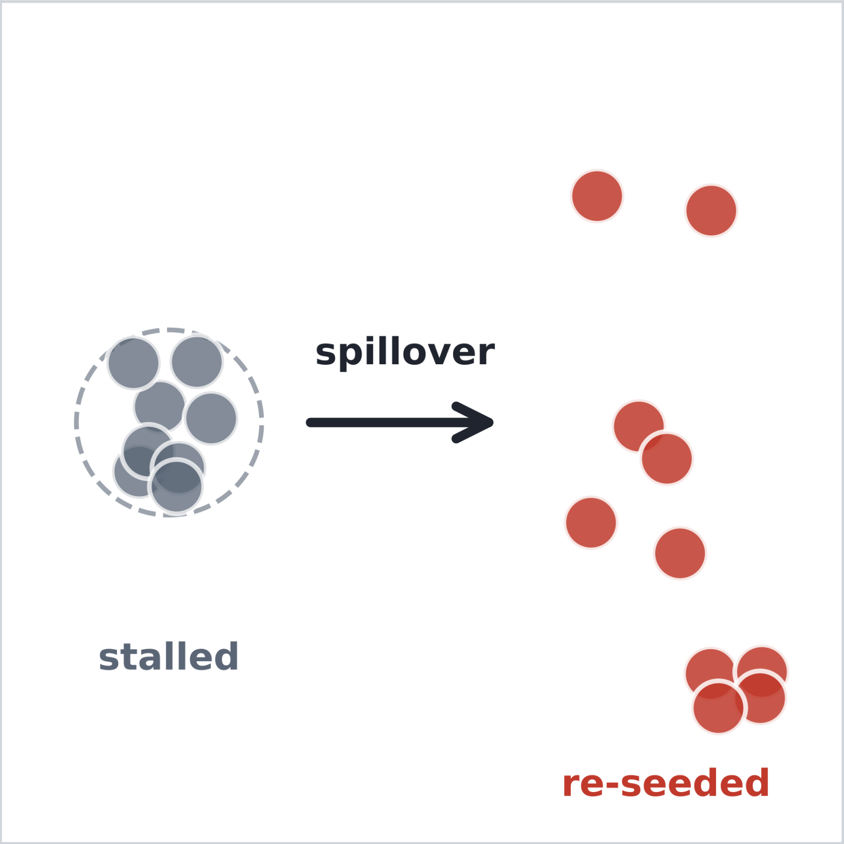
What happens

Drop the bottom quartile of the population.
For the rest, revert any child that lost to its parent.

Why it helps

Population improves monotonically — no regression.

Population mechanism — Spillover



Stalled cluster (left) re-seeded into wide scatter (right)

Spillover

Re-seed when stalled, switch basin if it persists

Activates when

After a long stagnation while the relative quality floor is still unmet.

What happens

First trigger: uniformly re-seed with a shrunken σ, keeping the best.
Second trigger: discard the best entirely and reset σ.

Why it helps

Escape stuck states. Switch basin when failure persists.

Step-size σ adaptation

Per-individual σ updates each generation from three signals

×1.10

On improvement

Child beats parent → explore wider.

×0.95

On stagnation

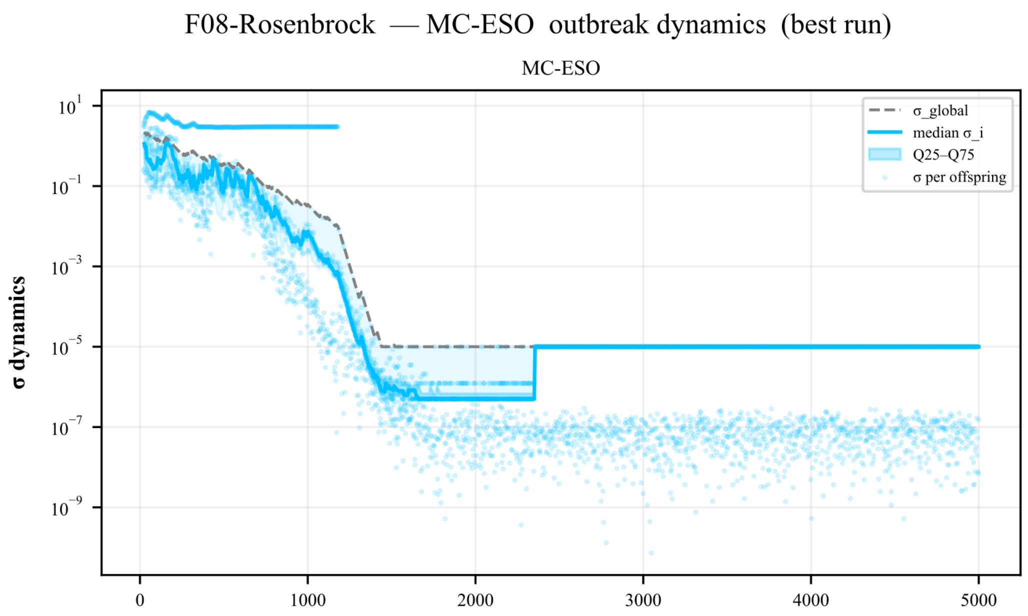
Parent wins → shrink and tighten.

×0.85

In drilling mode

Once σ becomes very small → drill fast, airborne off.

EVIDENCE



σ trajectory on F08-Rosenbrock

4

Experiments

BBOB benchmark — conditions and baselines

Experimental setup

Benchmark	BBOB-24 [15] + 11 classical = 35 functions
Dimension	$d = 2$, domain $[-5, 5]^2$
Budget	5,000 evals / run
Repetitions	30 runs / (function, method)
Baselines	CMA-ES [12], DE [13], PSO [14], SaVOA [2]
Metric	Success rate at precision 1e-k — fraction of runs reaching the target

Comparison methods (details in appendix)

MC-ESO	3 parallel channels (contact + droplet + airborne)
CMA-ES	Learns covariance C [12]
DE	DE/rand/1/bin [13]
PSO	Particle swarm [14]
SaVOA	Self-adaptive virus, single channel [2]

5

Results

Headline numbers and the hard cases

Mean SR across precision thresholds — all 5 methods

Best in each column bolded

Method	SR @ 1e-2	SR @ 1e-4	SR @ 1e-7	SR @ 1e-10
MC-ESO	**91.9%**	**90.9%**	**87.4%**	**84.3%**
DE	**92.4%**	88.7%	86.9%	**85.8%**
SaVOA	81.2%	75.7%	68.9%	67.5%
PSO	79.1%	75.3%	68.4%	56.9%
CMA-ES	64.5%	63.0%	62.2%	59.1%

vs. PSO	vs. CMA-ES	vs. SaVOA	vs. DE
30 W · 5 T · 0 L	**20 W · 12 T · 3 L**	**14 W · 17 T · 4 L**	**5 W · 22 T · 8 L**

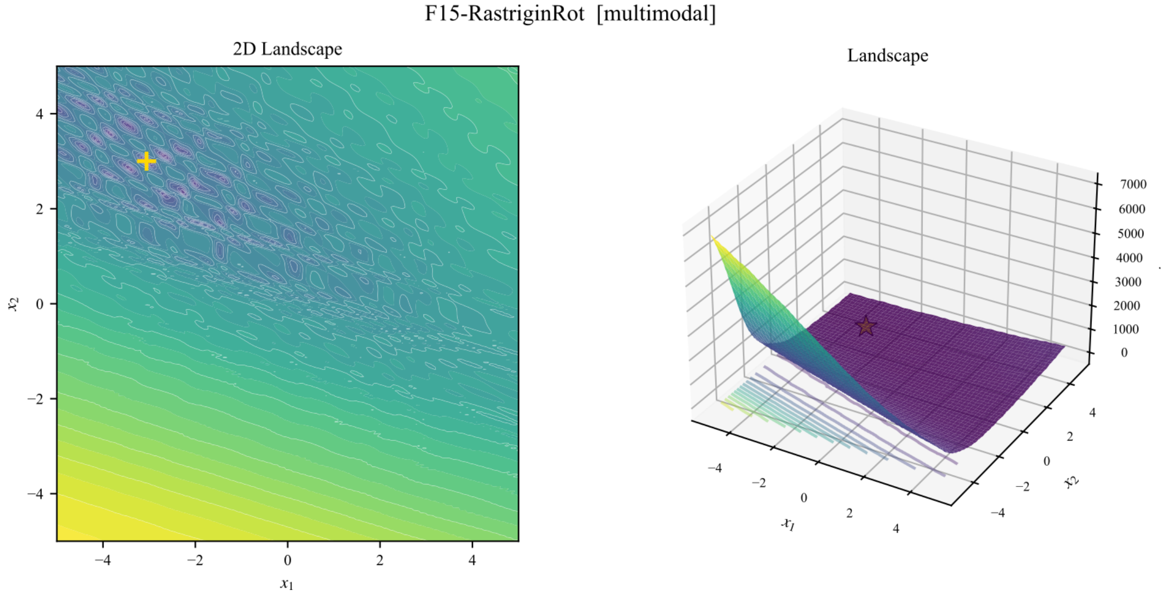
Wilcoxon signed-rank test [16] with 30 runs at the 5% level · Vargha–Delaney A12 [17] for effect size

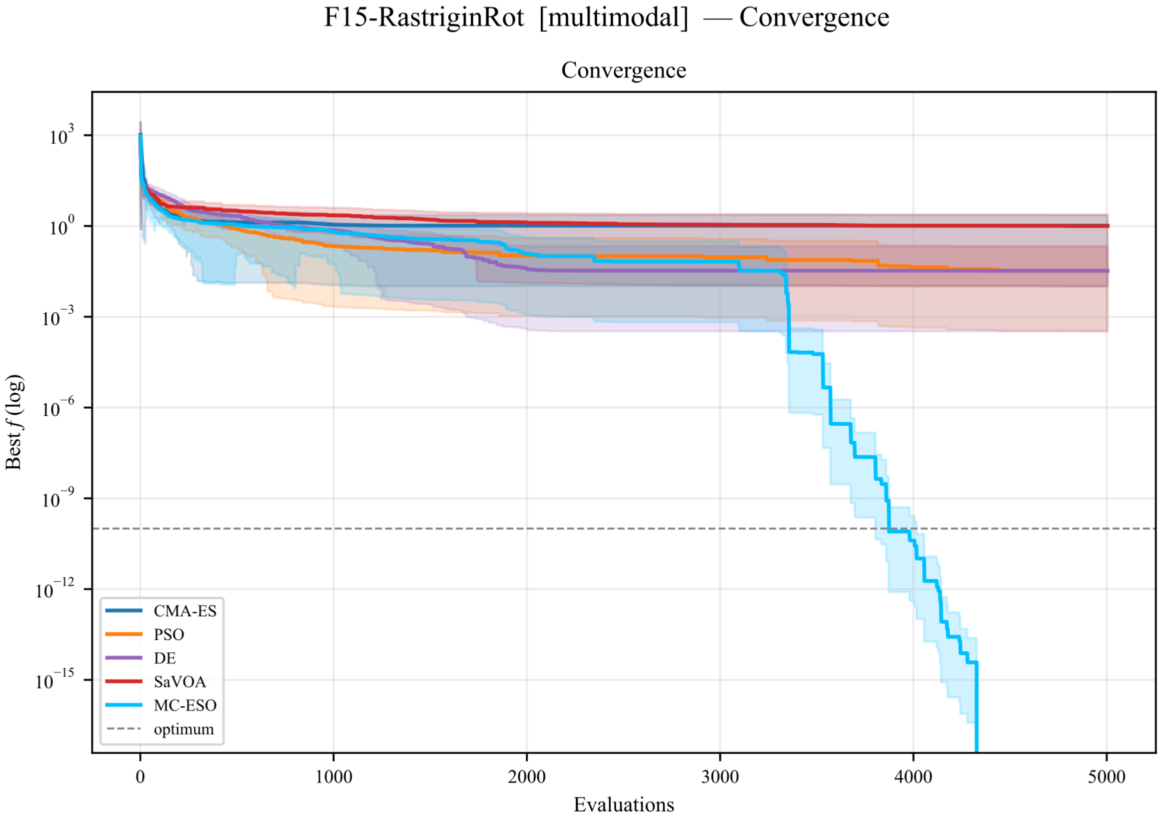
Where MC-ESO wins — landscape and convergence

MC-ESO solves 100% of runs while baselines collapse

F15 Rastrigin (rotated)

multimodal



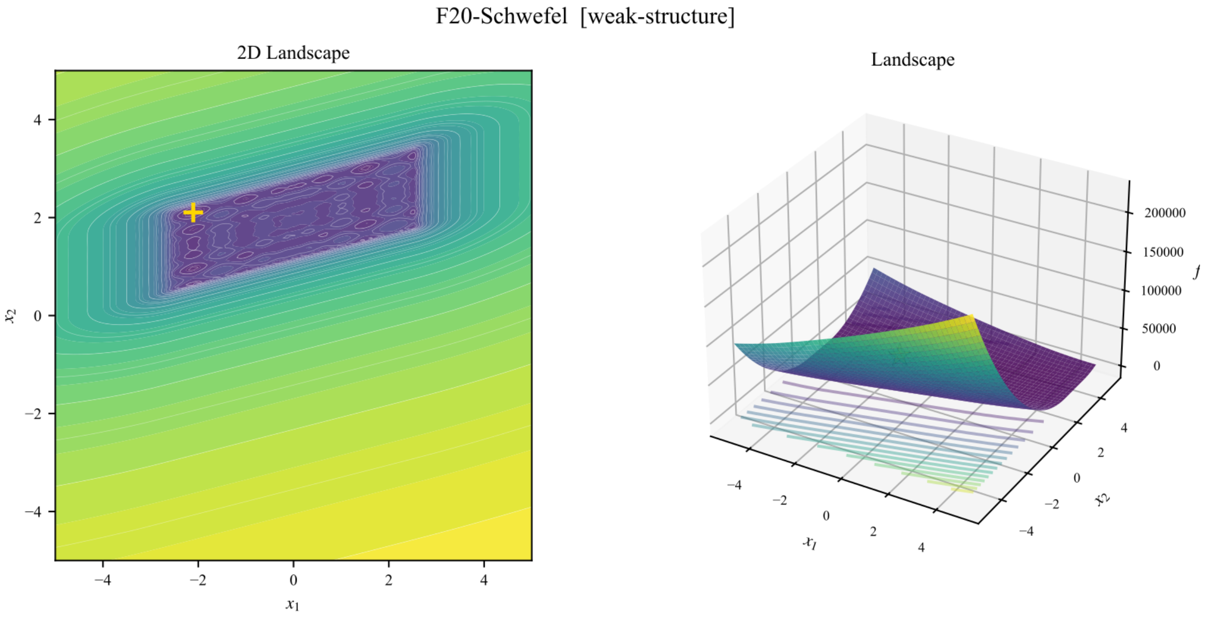


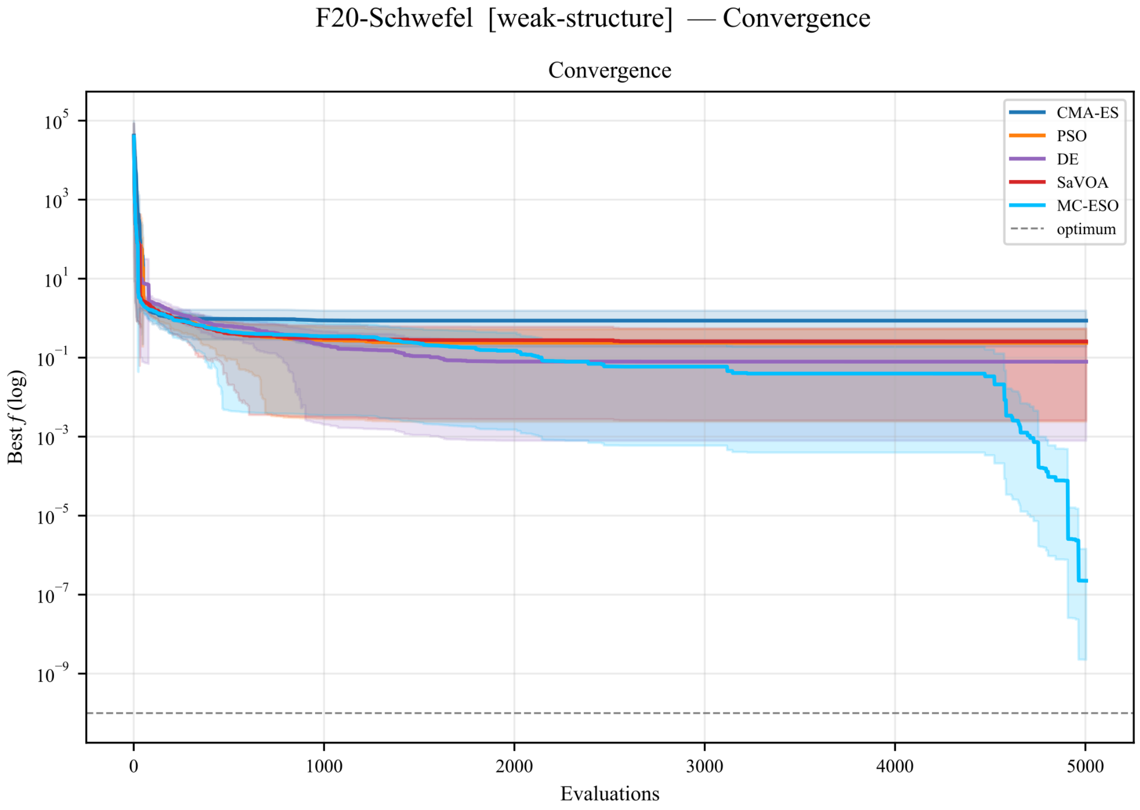
Where MC-ESO wins — landscape and convergence

MC-ESO solves 100% of runs while baselines collapse

F20 Schwefel

weak-structure





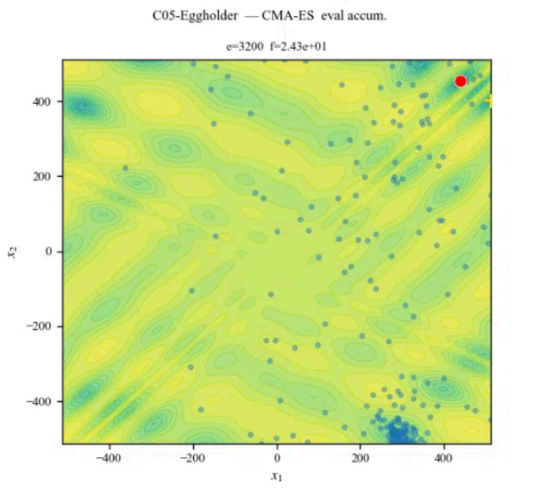
6 Discussion

Why it works, and where it loses

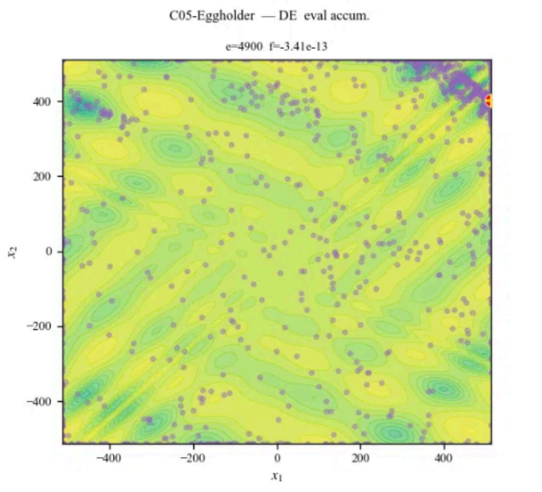
Why does it work — ① Exploration ability

Strain pool blocks premature commitment to one peak

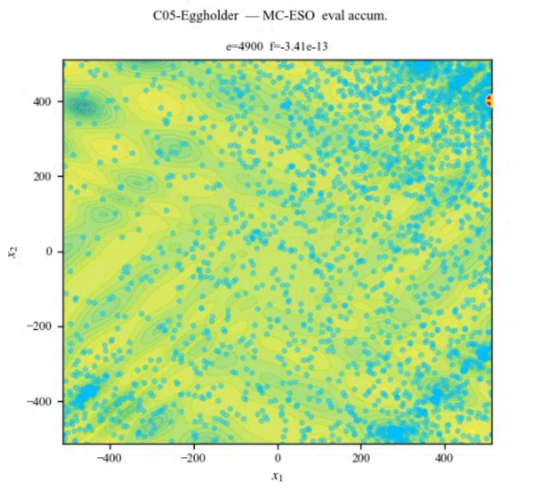
CMA-ES



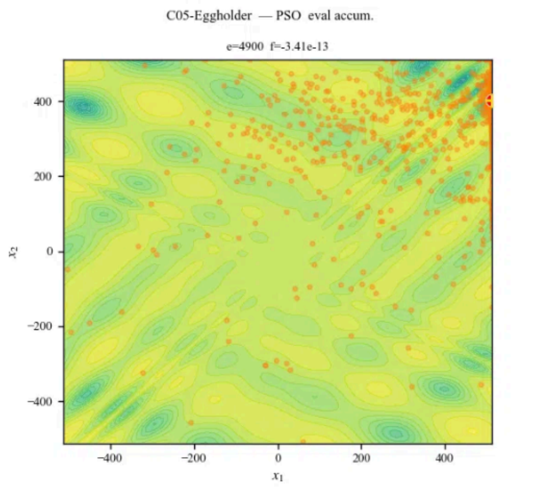
DE



MC-ESO



PSO



Mechanism behind it

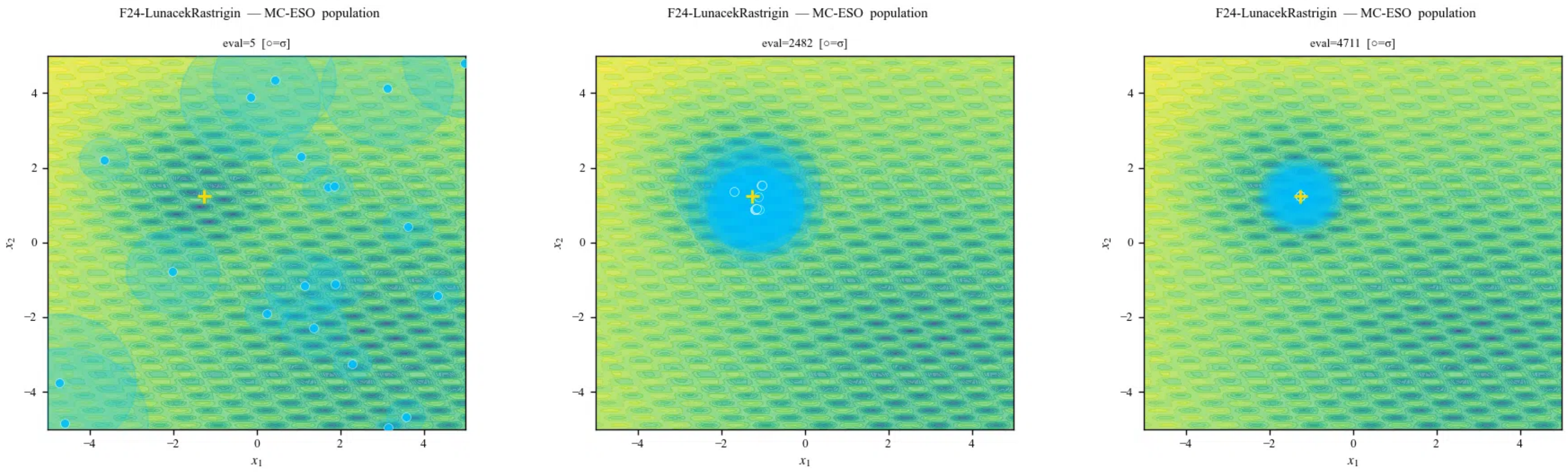
Up to 6 niched elites are kept spaced apart and 3 channels

Effect we observe

Droplet can pull toward ANY strain — no lock-in to the first attractor.

Why does it work — ② Adaptive disruption

Spillover escalates with each failure



Early → mid → late on F24 Lunacek bi-Rastrigin (best run)

Mechanism behind it

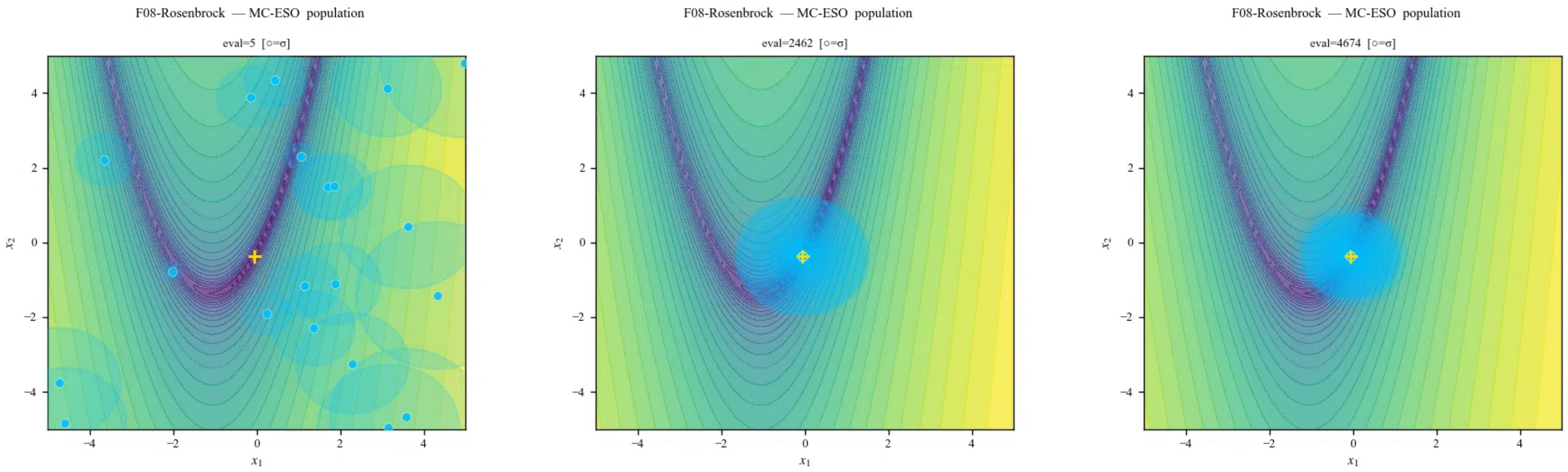
A stagnation counter is incremented each time the restart trigger fires without crossing the quality floor.

Effect we observe

On the first trigger: re-seed with a shrunken σ, keep best. On a second failure: discard the best entirely and reset σ.

Why does it work — ③ Implicit anisotropy

The droplet move automatically aligns with the population's spread



Early → mid → late on F08 Rosenbrock (best run)

Mechanism behind it

The difference vector is sampled from the current population, so its distribution naturally mirrors the population's covariance.

Effect we observe

Droplet moves stretch along the population's long axis — no explicit covariance matrix needs to be estimated.

Limitations

Two major issues that should be addressed

① DIMENSIONALITY

Only d = 2 reported

High-dim scaling unknown

- **Evaluated dimensions**
  Only two dimensions tested. Real black-box workloads are typically tens to hundreds.
- **Channels may not scale**
  Droplet's direction loses signal as the population shrinks relative to dimension. Covariance estimation also becomes noisy in high dim.
- **Strain pool may be too small**
  A fixed-fraction niche radius spans far less volume as dimensions grow, so six elites may no longer cover the search space.

Run at 5, 10, 20, and 40 dimensions

② NOVELTY / METHOD POLISH

Mechanisms reuse known ideas

Integration story is not yet sharp

- **Channels ≠ new operators**
  Contact = local Gaussian (ES), Droplet = current-to-best/1 (DE), Airborne = wide random — all known individually.
- **Strain pool ≈ standard niching**
  Crowding / clearing in EAs predates this work. The novelty is the link to the droplet channel, not niching itself.
- **Spillover ≈ escalating restart**
  IPOP/BIPOP-CMA-ES already escalates σ on stagnation. MC-ESO's streak rule needs a cleaner theoretical motivation.

Ablation per channel/mechanism and improve system

7

Conclusion

Summary

Conclusion

Idea

Multi-route reproduction (epidemic-inspired)

Mechanism

3 channels + 3 population mechanisms + adaptive σ

Result

90.9% SR @ 1e-4 — beats all 4 baselines

References

[1] Liang & Cuevas Juarez (2014). A Novel Metaheuristic for Continuous Optimization: Virus Optimization Algorithm. Engineering Optimization, 48(1).

[2] Liang & Cuevas Juarez (2020). A self-adaptive virus optimization algorithm for continuous space optimization. Soft Computing, 24.

[3] Martínez-Álvarez, Asencio-Cortés, Torres et al. (2020). Coronavirus Optimization Algorithm (CVOA): a bioinspired metaheuristic based on the COVID-19 propagation model. Big Data, 8(4).

[4] Oyelade, Ezugwu, Mohamed & Abualigah (2022). Ebola Optimization Search Algorithm (EOSA): a new nature-inspired metaheuristic. IEEE Access, 10.

[5] Hashemi, Joodaki, Joodaki et al. (2022). Anti Corona Virus Optimization Algorithm (ACVOA). Soft Computing.

[6] VDO authors (2025). Virus Diffusion Optimizer: a herpes life-cycle-inspired metaheuristic. arXiv:2510.24083.

[7] Qin, Huang & Suganthan (2009). Differential evolution algorithm with strategy adaptation for global numerical optimization. IEEE Trans. on Evolutionary Computation, 13(2).

[8] Mallipeddi, Suganthan, Pan & Tasgetiren (2011). Differential evolution algorithm with ensemble of parameters and mutation strategies. Applied Soft Computing, 11(2).

[9] Wang, Cai & Zhang (2011). Differential evolution with composite trial vector generation strategies and control parameters. IEEE Trans. on Evolutionary Computation, 15(1).

[10] Wu, Shen, Li, Chen, Lin & Suganthan (2018). Ensemble of differential evolution variants. Information Sciences, 423.

[11] Sun et al. (2023). RL-MDE: a reinforcement-learning-driven multi-strategy DE. Soft Computing / Applied Soft Computing.

[12] Hansen & Ostermeier (2001). Completely derandomized self-adaptation in evolution strategies (CMA-ES). Evolutionary Computation, 9(2).

[13] Storn & Price (1997). Differential evolution — a simple and efficient heuristic for global optimization over continuous spaces. J. of Global Optimization, 11(4).

[14] Kennedy & Eberhart (1995). Particle swarm optimization. Proc. of IEEE ICNN.

[15] Hansen, Auger, Ros, Finck & Pošík (2010). Comparing results of 31 algorithms from the black-box optimization benchmarking BBOB-2009. GECCO Workshop.

[16] Wilcoxon (1945). Individual comparisons by ranking methods. Biometrics Bulletin, 1(6).

[17] Vargha & Delaney (2000). A critique and improvement of the CL common-language effect-size statistics of McGraw and Wong. J. of Educational and Behavioral Statistics, 25(2).

Related work — multi-strategy / multi-operator algorithms

Each prior method picks ONE operator per offspring (or competes for budget)

Year	Method	Mechanism / how it generates samples
2009	**SaDE (Qin et al.) [7]**	Online learning, one strategy per offspring
2011	**EPSDE (Mallipeddi et al.) [8]**	Winners replicate from a competing (strategy, F, CR) pool
2011	**CoDE (Wang et al.) [9]**	Generate 3 trial vectors, take the best
2018	**EDEV (Wu et al.) [10]**	3 subpops (JADE/CoDE/EPSDE) compete for compute budget
2023	**RL-MDE [11]**	RL picks the mutation strategy per subpopulation

Deep-dive — EDEV Ensemble of DE Variants

Info. Sciences, 2018 (Wu et al.) [10] · Structurally closest multi-strategy method — three subpopulations, three operators

1

Three indicator subpops

Equal subpops run JADE / CoDE / EPSDE.

2

Reward subpop competes

Larger 4th subpop reassigned to the winner.

3

Information sharing

Re-partition each gen — good moves migrate.

Figure placeholder

EDEV 4-subpop architecture diagram
(Wu et al., Info. Sciences 2018, Fig. 1)

→ drop a screenshot at
figures/screenshots/edev.png

Appendix — SaVOA (Self-adaptive Virus Optimization)

Closest prior work — virus-inspired, single channel [2]

Inspiration	Viral infection + self-replication
Reproduction	Single Gaussian around parent (contact only)
σ update	Self-adaptive per agent
Population	Generational replacement, no niching, no restart
vs. MC-ESO	• 1 channel → 3 • + 3 population mechanisms

Appendix — Other baselines

DE [13]

Differential Evolution — DE/rand/1/bin

- Mutation: take a random individual and add a scaled difference of two peers
- Strong general-purpose baseline

CMA-ES [12]

Covariance Matrix Adaptation ES

- Learns sampling covariance
- Strong on rotated landscapes
- Weak on multimodal functions

PSO [14]

Particle Swarm Optimization

- Particles track own + swarm best
- Strong on smooth landscapes
- Stagnates on narrow valleys